CRL Conditional Retro-Leverage

The Early Stop-Out Problem in Professional Leveraged Trading

A conditional leverage approach designed to reduce early account losses and extend client relationship duration

PUBLIC INFORMATION BRIEF. FOR PROFESSIONAL BROKER DISCUSSION ONLY.

This document describes the structural early stop-out problem and the CRL approach at a high level. Any figures or examples are illustrative, based on stated assumptions, and should not be interpreted as forecasts, investment returns, client performance claims, or guarantees of live trading outcomes. Broker-specific validation should be conducted using the broker's own instruments, client segments, funding curves, execution conditions and risk parameters.

Public Overview for Professional Counterparties
Version 1.2

CRL Technologies Inc.
info@crl-technologies.com

1. The Problem

Brokers serving professional traders observe a persistent pattern: a significant portion of new clients experience account depletion within the first 30 days of activity. In many cases the underlying trade direction eventually proves correct, but early adverse volatility triggers margin events before the thesis can materialise.

This is a structural feature of traditional leverage. Exposure is applied from the moment of position inception, creating an asymmetry between immediate downside vulnerability and deferred upside realisation.

Economic Impact

The cost to brokers is measurable. Each early stop-out represents unrecovered client acquisition cost, foregone lifetime revenue from trading activity, and potential reputational friction in concentrated client networks.

The magnitude varies by broker, but the pattern is consistent: early account losses disproportionately affect clients who may have been profitable over a longer relationship horizon.

2. The CRL Approach

Conditional Retro-Leverage (CRL) is a calculation methodology that brokers implement within their own products. Its defining property is simple to state: leverage does not amplify exposures that the market has not yet confirmed.

Before confirmation, the position is not amplified by leverage, so early adverse volatility does not produce leveraged drawdown. Leverage applies only after the market has moved in the direction of the trade beyond a defined threshold. The mechanism that realises this property is the means, not the point.

Two consequences follow. First, the principal observable effect is a reduction in early account depletion driven by leveraged exposure taken before directional confirmation. Second, CRL does not improve the trader's ability to predict market direction; it changes how incorrect timing is treated relative to confirmed directional moves.

Consistent with no-arbitrage, at fair pricing CRL does not create excess expected return. It reshapes the distribution of outcomes, in particular the timing and concentration of risk, rather than adding expected value.

CRL provides the calculation layer. The broker remains the sole issuer, counterparty and distributor of any product incorporating it. CRL does not execute orders, clear trades, hold client funds, or provide investment advice.

CRL does not eliminate directional risk, does not guarantee client retention, does not prevent losses, and is not capital protection except where a specific product wrapper expressly provides it.

3. Validation

Independent validation has been conducted through a multi-year historical backtest on real market data and a large-scale Monte Carlo simulation. The principal observed effect is a reduction in early account depletion, relative to constant-leverage benchmarks and under stated assumptions, together with a reshaping of the outcome distribution, driven by the absence of leveraged exposure during the pre-confirmation phase rather than by improved trade selection or predictive accuracy.

Under fair pricing assumptions, CRL does not generate excess expected value. Its function is to redistribute risk and return over time, not to add expected return. From a broker perspective, CRL does not eliminate risk; it transforms its timing and concentration, which has direct implications for hedging and exposure management.

Interpretation

Simulated results rest on simplifying assumptions and are not predictive of live outcomes. Real-world performance depends on execution, slippage, market microstructure, implementation specifics, and client behaviour. No simulated metric should be interpreted as predictive of live trading outcomes.

Quantitative dossier

The full quantitative dossier, including methodology, configurations and results, is available to qualified counterparties under a non-disclosure agreement. For brokers requiring deeper validation, the framework can be re-run on the broker's own instruments, feeds, funding curves and risk metrics through a joint study, producing a validation dossier co-developed with the broker's risk and product teams.

4. Technical Integration

CRL is delivered as a white-label API designed for integration with existing broker trading infrastructure.

Architecture

- API specification: REST with OpenAPI 3.0 documentation. SDK support for Python, TypeScript, Java, C# and Go.

- Authentication: HMAC-SHA256 request signing over TLS 1.2 or 1.3. Mutual TLS can be designed on request.
- Performance: on the current single-node deployment, a signed API round-trip averages about 58 ms, with p99 up to about 250 ms end-to-end.

- Audit trail: cryptographically signed, tamper-evident logs designed for regulatory review.

Regulatory Considerations

CRL is designed for professional clients only, under the classification frameworks of MiFID II and equivalent regimes. The system provides supporting documentation packages intended to be compatible with product governance, best execution, and transaction reporting requirements. Final regulatory assessment remains the responsibility of the integrating broker and their legal and compliance teams.

5. Pilot Framework

For brokers interested in evaluating CRL, we propose a structured pilot programme to validate the approach against their specific client base.

Proposed Structure

Duration	90 days
Client segment	100 to 200 professional clients, structured as A/B test
Asset classes	FX majors, equity indices, or commodities (to be agreed)
Success metrics	Early stop-out rate, 30-day retention, client feedback

Joint Validation Option

For institutions requiring deeper validation, a joint study re-runs the CRL framework on the institution's own instruments, feeds, funding curves and risk metrics. This produces a validation dossier co-developed with the institution's risk and product teams.

6. Next Steps

If reducing early liquidation driven by timing rather than direction is a priority for your business, we would welcome the opportunity to discuss further.

1. **Discovery call:** a short conversation to understand your current metrics and discuss whether CRL is a suitable fit.
2. **Technical review:** an API walkthrough with your integration team to assess implementation feasibility.
3. **Pilot proposal:** a customised pilot structure based on your client segments and asset classes.

The initial discussion is intended to determine whether CRL is relevant to the broker's product, risk and client-retention priorities.

Contact

Andrea Galié
Founder
andrea@crl-technologies.com

CRL Technologies Inc. | Delaware C-Corporation
This overview is intended for professional broker decision-makers. CRL products are designed exclusively for MiFID II Professional Clients and Eligible Counterparties and are not offered to retail clients.